4B
AQUEOUS SOLUTIONS

UNIT 01: FOUNDATIONAL CHEMISTRY SKILLS

CHEMISTRY 1310

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LEARNING OBJECTIVES

CHAPTER 4

1. Write complete and balanced chemical equations, based on either chemical symbols or word descriptions, including appropriate phases.
2. Recognize and describe the five basic types of chemical reactions.
3. List and describe the driving forces for reactions in aqueous solution.
4. Describe and write equations representing the dissociation of ionic compounds upon dissolution in water.
5. Define and apply the terms strong electrolyte, weak electrolyte, and nonelectrolyte.
6. Identify compounds as being strong electrolytes, weak electrolytes, or nonelectrolytes from chemical formulas.
7. Apply solubility guidelines to predict the formation of a precipitate in reactions of ionic compounds in aqueous solution.
8. Write and interpret ionic and net ionic equations for precipitation reactions.

Compounds in Aqueous Solution

4.3

Aqueous solutions

- Result when a substance is dissolved in water.

Components of a solution:

- **Solvent**: Component present in greater amounts (usually much greater amounts); the major component
  - Water is the solvent in aqueous solutions.
- **Solute**: Component present in lesser amounts; the minor component

In all solutions, the solute dissolves in the solvent.

Compounds in Aqueous Solution

4.3

Soluble

- Substances dissolve easily and to a large extent in a specified solvent.

Insoluble

- Substances do not dissolve to any measurable extent in a specified solvent.

Solubility rules are a quick way to see what substances are soluble or insoluble.

Compounds in Aqueous Solution

4.3

When ionic compounds dissolve, they dissociate or break apart into their constituent ions (cations and anions).

- The associated reaction is called a dissociation reaction.

$$NaCl(s) \rightarrow Na^+(aq) + Cl^-(aq)$$

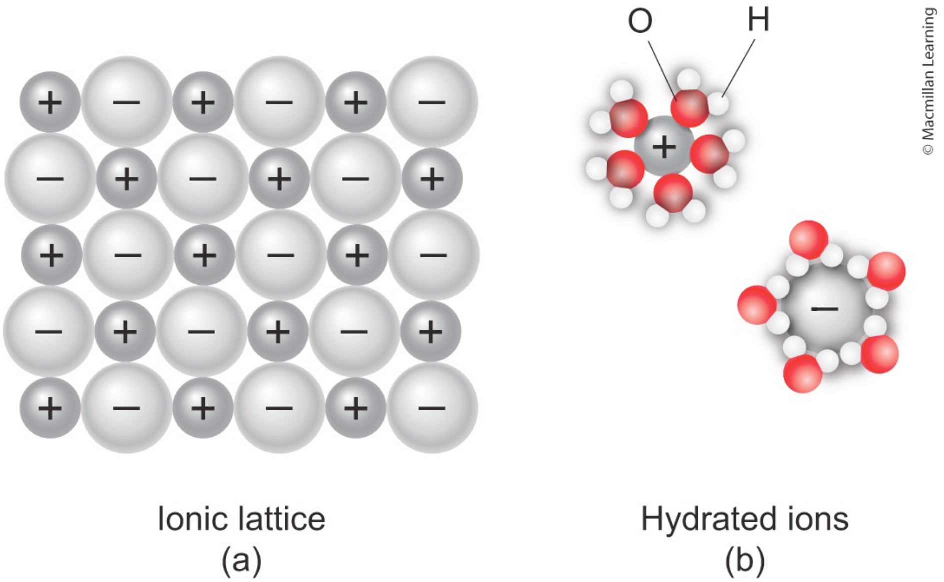


FIGURE 4.15

Compounds in Aqueous Solution

4.3

Electrolytes: Substances that dissolve in water to produce a conductive solution.

- Strong electrolytes: Dissolves and dissociate completely (soluble ionic compounds, strong acids, and strong bases)
- Weak electrolytes: Dissolves but dissociate only a little bit (insoluble ionic compounds, weak acids, and weak bases)

Nonelectrolytes: Substance that dissolve but do not dissociate (covalent molecules)

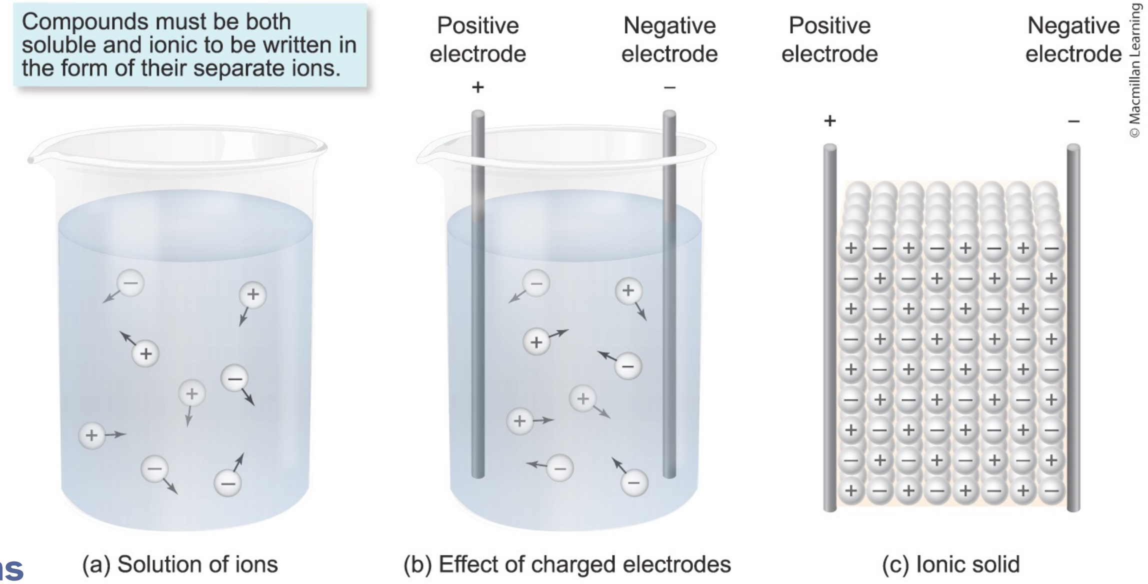


FIGURE 4.16 Mobility of ions

Compounds in Aqueous Solution

4.3

Figure 4.17 of your book gives you an opportunity to explore different compounds to determine if they are strong electrolytes, weak electrolytes, or nonelectrolytes.

TABLE 4.4
Electrolytic Properties of Various Types of Compounds

Solution Type	Solute Compound Type	Examples
Strong electrolytes	Soluble ionic compounds (salts)	$NaCl(aq)$, $K_2SO_4(aq)$
	Strong acids	$HCl(aq)$, $HNO_3(aq)$
	Strong bases	$NaOH(aq)$, $KOH(aq)$
Weak electrolytes	Insoluble ionic compounds (salts)	*See Section 4.4*
	Weak acids	$HF(aq)$, $HNO_2(aq)$, $H_3PO_4(aq)$
	Weak bases	$NH_3(aq)$, $CH_3NH_2(aq)$
Nonelectrolytes	Covalent molecules (most)	$C_6H_{12}O_6(aq)$, sugars

Compounds in Aqueous Solution

4.3

Strong Acids: Form solutions that conduct electricity very well and are strong electrolytes

- There are only *seven* strong acids you should be familiar with.
- ALL other acids are weak acids by definition.

TABLE 4.3

Strong Acid Name	Formula	Ions in Aqueous Solution	
Hydrochloric acid	HCl	H^+	Cl^-
Hydrobromic acid	HBr	H^+	Br^-
Hydroiodic acid	HI	H^+	I^-
Nitric acid	HNO_3	H^+	NO_3^-
Perchloric acid	$HClO_4$	H^+	ClO_4^-
Chloric acid[1]	$HClO_3$	H^+	ClO_3^-
Sulfuric acid[2]	H_2SO_4	H^+	HSO_4^-

Notes:

1. Some sources consider chloric acid to be a weak acid.
2. Only one of the H atoms completely dissociates. In other words, H_2SO_4 is a strong acid but HSO_4^- is a weak acid.

Weak Acids: Form solutions that conduct electricity only slightly and are weak electrolytes

Compounds in Aqueous Solution

4.3

Strong Bases: Form solutions that conduct electricity very well and are strong electrolytes

- Ionic compounds containing hydroxide ions (OH^-) that dissociate *completely* in water.
- Only soluble hydroxide compounds are strong bases. You should be familiar with these.
- ALL other bases are weak bases by definition.

Strong Base Name	Formula	Ions in Aqueous Solution	
Lithium hydroxide	$LiOH$	Li^+	OH^-
Sodium hydroxide	$NaOH$	Na^+	OH^-
Potassium hydroxide	KOH	K^+	OH^-
Calcium hydroxide*	$Ca(OH)_2$	Ca^{2+}	OH^-, OH^-
Strontium hydroxide*	$Sr(OH)_2$	Sa^{2+}	OH^-, OH^-
Barium hydroxide*	$Ba(OH)_2$	Ba^{2+}	OH^-, OH^-

* Note: When $Ca(OH)_2$, $Sr(OH)_2$, and $Ba(OH)_2$ dissociate in water, they yield <u>two</u> OH^- anions per unit of strong base.

Weak Bases: Form solutions that conduct electricity only slightly and are weak electrolytes

- Molecular compounds (e.g. ammonia, NH_3) that react with water to a small extent to produce OH^- ions in solution.

Precipitation Reactions

4.4

Ionic compounds that readily dissolve in water are said to be soluble, whereas ionic compounds that do not readily dissolve in water are said to be insoluble.

- Even *insoluble* ionic compounds dissolve to some degree, but the amount is so small that it can be ignored for our purposes.

SOLUBILITY GUIDELINES

1. Compounds of group 1 elements (Li^+, Na^+, K^+, Rb^+, Cs^+, and Fr^+) and ammonium (NH_4^+) are soluble.
2. Nitrates (NO_3^-), chlorates (ClO_3^-), perchlorates (ClO_4^-), and acetates ($C_2H_3O_2^-$) are soluble.
3. Chlorides (Cl^-), bromides (Br^-), and iodides (I^-) are soluble, except for those of Ag^+, Pb^{2+}, and Hg_2^{2+}.
4. Except for compounds of the cations in guideline 1, carbonates (CO_3^{2-}), sulfites (SO_3^{2-}), phosphates (PO_4^{3-}), and chromates (CrO_4^{2-}) are insoluble.
5. With the exception of guideline 1 and the barium ion (Ba^{2+}), hydroxides (OH^-) and sulfides (S^{2-}) are insoluble.
6. With the exception of guideline 2, silver (Ag^+), mercury (Hg_2^{2+}), and lead (Pb^{2+}) salts are insoluble.
7. With the exception of compounds of calcium (Ca^{2+}), strontium (Sr^{2+}), barium (Ba^{2+}), and the ions listed in guideline 6, all sulfates are soluble.

IN-CLASS QUESTION 1

Learning Objective(s): 4.5, 4.6

Aqueous solutions of which substances conduct electricity *strongly*?

Select all that apply.

- ❑ sucrose ($C_{12}H_{22}O_{11}$)
- ❑ KOH
- ❑ sodium phosphate
- ❑ copper(II) carbonate
- ❑ hydrobromic acid
- ❑ HNO_2
- ❑ CH_3CH_2COOH
- ❑ $HClO_4$

IN-CLASS QUESTION 2

Learning Objective(s): 4.6

Which statement is **true** regarding an aqueous solution of barium hydroxide?

- ❑ The solution contains many intact $Ba(OH)_2$ formula units.
- ❑ The solution contains equal amounts of barium ions and hydroxide ions and very few (if any) $Ba(OH)_2$ formula units.
- ❑ The solution contains equal amounts of barium ions, hydroxide ions, and $Ba(OH)_2$ formula units.
- ❑ The solution contains equal amounts of barium ions and $Ba(OH)_2$ formula units. It contains twice as many hydroxide ions as barium ions and $Ba(OH)_2$ formula units.
- ❑ The solution contains twice as many hydroxide ions as barium ions, and it contains very few $Ba(OH)_2$ formula units.

Precipitation Reactions

4.4

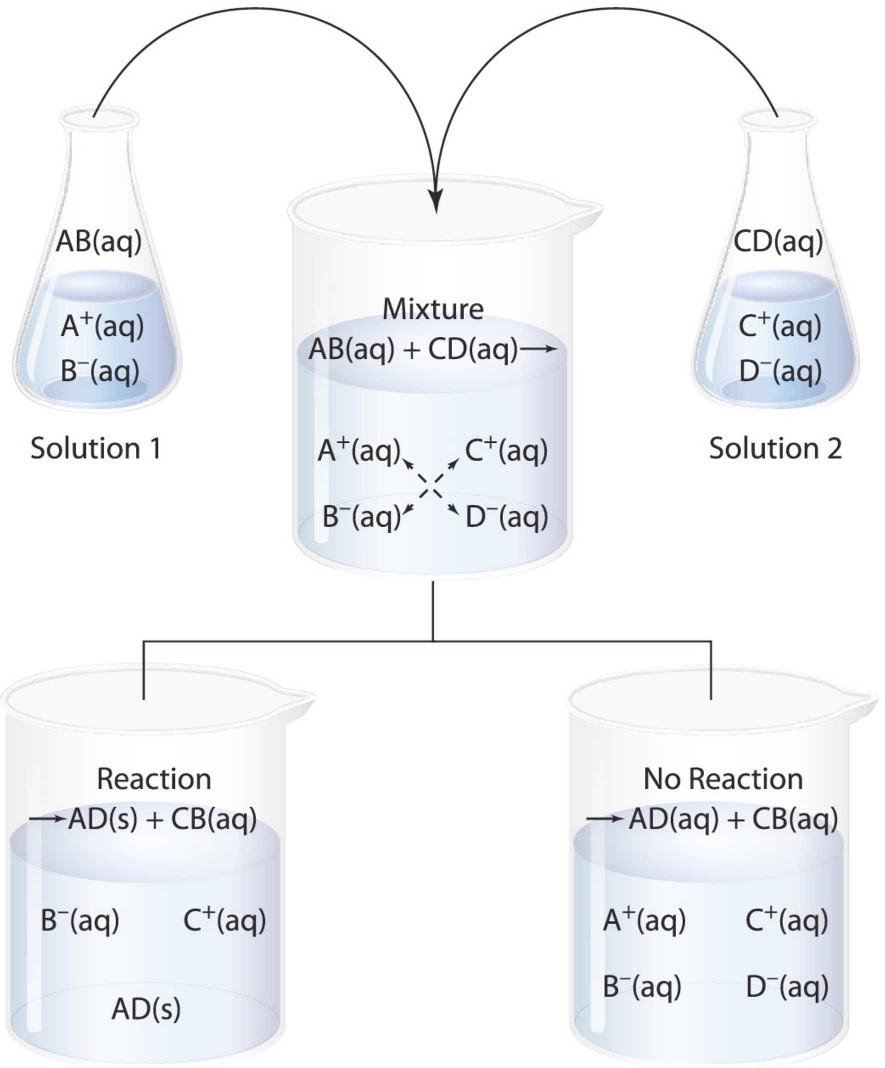
The mixture of aqueous solutions of two soluble ionic compounds results in a solution containing four ions.

- *These are the double-replacement reactions we examined earlier in the chapter.*

Precipitation reactions do not always occur when two aqueous solutions are mixed.

- Only ***insoluble*** compounds form precipitates.
- If both products are soluble, then NO REACTION occurs.

FIGURE 4.19 Two Solutions of Ionic Compounds



EXAMPLE PROBLEM 1

Learning Objective(s): 4.7

The following pairs of aqueous solutions are mixed together. For each pair, determine whether a precipitate forms or if there is no reaction.

- K_2CO_3 and $NiCl_2$
- Li_2SO_4 and $BaCl_2$
- NH_4NO_3 and K_3PO_4
- $AgNO_3$ and $NaCl$
- $CaBr_2$ and Na_2SO_4

Precipitation Reactions

4.4

Aqueous chemical reactions can be written as a total equations (often called molecular equations).

- The complete formula for each ionic compound is shown.

Complete ionic equations

- Emphasize dissociation of reactants and products by writing all **strong electrolytes** in their dissociated forms (dissociated ions).
- REMEMBER: Only aqueous ionic substances dissociate. Gases, liquids, solids, and covalent aqueous species NEVER dissociate.

Net ionic equations

- Simplify the total ionic equation to show only the components that undergo direct change in the reaction by omitting the spectator ions.
- Spectator ions: Ions that appear *exactly the same* on both sides of the chemical equation.

IN-CLASS QUESTION 3

Learning Objective(s): 4.7, 4.8

A precipitate forms when aqueous solutions of aluminum sulfate and barium chloride are mixed.

Write the molecular and net ionic equations for the reaction.

EXAMPLE PROBLEM 2

Learning Objective(s): 4.7

Predict the products of the reaction of aqueous solutions of cadmium(II) sulfate and potassium sulfide.

Write a balanced molecular equation (include phases) and identify all spectator ions.